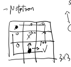
RL Intro

Friday, January 16, 2026 21:57

- Notation



state value

$s \in S$, $a \in A$, $r \in R$

$\boxed{V(s)}$, $\boxed{q(s,a)}$ → action value

$\pi(a|s)$ → policy

$V(s) = 0.5\times 100 + 0.5\times -100 = 0$

$q(s, a=\text{right}) = 100$

- MDP $s \xrightarrow[r]{a} s' \xrightarrow{a'} s''$ discount factor

- Bellman equation $V_\pi(s) = E(r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \dots + \gamma^{T-t} r_T \mid S = S_t)$

- DP $= \sum_a \pi(a|s) \sum_{s'} p(s', r|s,a)[r + V_\pi(s')]$ → Bellman equation ($\to q_\pi(s,a)$)

$\pi^* = \arg\max_\pi V_\pi(s)$

$V_\pi(s) < q_\pi(s,a)$

$V_{\pi^*}(s) = \max_a q_{\pi^*}(s,a) = \max_a \sum_{s'} p(s', r|s,a)[r + \gamma V_{\pi^*}(s')]$
$q_{\pi^*}(s,a) = \sum_{s'} p(s', r|s,a)[r + \gamma \max_{a'} q_{\pi^*}(s', a')]$
} Bellman optimality equation

- model-based method < PI, VI env response: $p(s'|s,a)$, r
- model-free method < MC, TD

- policy iteration < PE (estimation), PI (improvement)

- initial π, $V(s)$ for $s \in S$

Loop
- for $s \in S$:

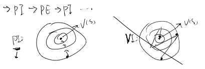
PE {
$V(s) = \sum_a \pi(a|s) \sum_{s'} p(s', r|s,a)[r + \gamma V(s')]$
$|V_{old}(s) - V(s)| < \varepsilon$ → $V(s)$ is convergent

- for $s \in S$:

PI {
$a^* = \arg\max_a q(s,a)$
if $\pi_{a=a^*}(a|s) \neq 1$ → π not optimal → $\pi_{a=a^*}(a|s) = 1$ else $\pi_{a\neq a^*}(a|s) = 0$

- if π is not optimal, go to PE again

PE → PI → PE → PI …



- value iteration
- initialize π, $V(s)$ for $s \in S$

Loop
- for $s \in S$:
  - $V(s) = \arg\max_{a^*} q(s,a)$ → value update
  - if $\pi_{a=a^*}(a|s) \neq 1$, π is not optimal, → $\pi_{a=a^*}(a|s) = 1$ else $\pi_{a\neq a^*}(a|s) = 0$ → policy update
  - $|V_{old}(s) - V(s)| < \varepsilon$ → V is convergent
- output π, V

- MC

① × env response $p(s'|s,a)$ [generate experience + avg]

② $S\uparrow$, $V(s)$ cost $\uparrow$,

$p(S_t)$, randomly assign start port (S_0, a_0) — MC with start port

π < exploitation, (greedy action) → $\varepsilon = 0.1$
exploration (ε-greedy
$a^* = \arg\max_a q(s,a)$
$\pi_{a=a^*}(a|s) = 1 - \varepsilon + \frac{\varepsilon}{|A(s)|}$
$\pi_{a\neq a^*}(a|s) = \frac{\varepsilon}{|A(s)|}$)

$S_0 \xrightarrow[r_1]{a_0} S_1 \xrightarrow[r_2]{a_1} S_2 \cdots S_{T-1} \xrightarrow[r_T]{a_{T-1}} S_T$ experience / episode / trajectory

✓ $q(s,a)$ × $V(s)$, $q(S_t, a_t) = E[r_{t+1} + \dots + \gamma^{T-t} r_T \mid S = S_t, a = a_t] = E[G_t \mid S = S_t, a = a_t]$

version 1
① MC w start port
- initialize π, $q(s,a)$, returns(s,a) empty list for $s \in S$
- for episode k in 1, 2 … n:
  - randomly select (S_0, a_0)
  - $\pi \to S_0, a_0, r_1, S_1, a_1, \dots r_T, S_T$
  - for t in $T-1, T-2, \dots 0$:
    - $G_t^k \to G_{t-1}^k + \gamma r_t$
    - append G_t^k into returns(S_t, a_t) list
    - $q^k(S_t, a_t) = $ avg returns(S_t, a_t)
    - $a^* = \arg\max_a q(S_t, a_t)$
    - $\pi_{a=a^*}(a|s) = 1$ and $\pi_{a\neq a^*}(a|s) = 0$

只考虑 first-visited

② ε-greedy method
由 $\pi \to S_0, a_0, r_1 \dots r_T, S_T$
$\pi_{a=a^*}(a|s) = 1 - \varepsilon + \frac{\varepsilon}{|A(s)|}$
and $\pi_{a\neq a^*}(a|s) = \frac{\varepsilon}{|A(s)|}$

- incremental implementation

Same (S_t, a_t) $G_t^1, G_t^2, \dots G_t^{\text{n-1}}$

$$q^n(S_t, a_t) = \frac{\sum^{n-1} G_t^k}{n-1} = \frac{\sum^{n-1} W^k G_t^k}{\sum^{n-1} W^k}$$

$$q^{n+1}(s,a) = \frac{\sum^n G_t^k W^k}{\sum^n W^k} \to q^{n+1}(s,a) = \frac{q^n(s,a)\sum^{n-1} W^k + G^n W^n}{\sum^n W^k} = \frac{q^n \sum^{n-1} W^k + G^n W^n + q^n W^n - q^n W^n}{\sum^n W^k}$$

$$= \frac{q^n \sum_{k=1}^{n} W^k + W^n(G^n - q^n)}{\sum_{k=1}^{n} W^k} = q^n + \frac{W^n}{\sum_{k=1}^{n} W^k}(G^n - q^n)$$

α $\theta^{n+1} = \theta^n + \alpha \nabla J(\theta)$

version 2 (w incremental implementation)
- initialize π, $q(s,a)$, returns(s,a) empty list for $s \in S$
- for episode k in 1, 2, … n
  - $\pi \to S_0, a_0, r_1 \dots S_T, a_T$
  - for t in $T-1, T-2 \dots 1$:
    - $G_t^k \to G_{t-1}^k + \gamma r_t$
    - $q^{k+1}(S_t, a_t) = q^k + \frac{W^k}{\sum W^k}(G_t^k - q^k(S_t, a_t))$

- TD learning

$q(S_t, a_t) \leftarrow q(S_t, a_t) + \alpha[G_t - q(S_t, a_t)]$ ($G_t \to r_{t+1} + \gamma q(S_{t+1}, a_{t+1})$)

$V(S_t) \leftarrow V(S_t) + \alpha[G_t - V(S_t)]$ ($G_t \to r_{t+1} + \gamma V(S_{t+1})$)

$V(S_t) \leftarrow V(S_t) + \alpha[\underbrace{r_{t+1} + \gamma V(S_{t+1})}_{\text{TD target}} - V(S_t)]$ 1-step learning: TD(0) → TD error

- proof:

$V(S_t) = E[G_t \mid S = S_t] = E[r_{t+1} + \gamma G_{t+1} \mid S = S_t]$
$= E[r_{t+1} + \gamma V(S_{t+1}) \mid S = S_t]$ → comb of MC and DP

- formulation of TD error:

$G_t - V(S_t) = r_{t+1} + \gamma G_{t+1} - V(S_t) + \gamma V(S_{t+1}) - \gamma V(S_{t+1})$
$= \delta_t + \gamma(G_{t+1} - V(S_{t+1})) = \delta_t + \gamma\delta_{t+1} + \gamma(G_{t+2} - V(S_{t+2})) = \dots = \sum_{k=t}^{T-1} \gamma^{k-t} \delta_k$

where $\delta_t = r_{t+1} + \gamma G_{t+1} - V(S_t)$

- property

TD learning variance ↓, bias ↑ compared to MC

$S_t, S_{t+1}, a_{t+1}, r_{t+1}$ → fewer var

- value func approx

$V(s) \to \hat{V}(s, w)$, 同理 $q(s,a) \to \hat{q}(s,a,w)$

obj func $J(w) = \arg\min_w E[(V(s) - \hat{V}(s,w))^2]$
$= \arg\min_w \frac{1}{2} E[(V(s) - \hat{V}(s,w))]^2$

$\frac{\partial J(w)}{\partial w} = -(V(s) - \hat{V}(s,w))\nabla\hat{V}(s,w)$ → SGD

$W_{t+1} = W_t - \alpha\frac{\partial J(w)}{\partial w} = W_t + \alpha(V(S_t) - \hat{V}(S_t, w_t))\nabla\hat{V}(S_t, w_t)$, 同理 for $q(s,a,w)$

$W_{t+1} = W_t + \alpha(q(S_t, a_t) - \hat{q}(S_t, a_t, w_t))\nabla\hat{q}(S_t, a_t, w_t)$

- MC $V(s)$ → replace by $G(t)$
- TD(0) $V(s)$ → replace $r_{t+1} + \gamma\hat{V}(S_{t+1}, w)$ → semi-gradient, 同理 for $q(s,a)$ → replace $r_{t+1} + \gamma\hat{q}(S_{t+1}, a_{t+1}, w_{t+1})$

- feature construction

$\hat{V}(s) / \hat{q}(s,a,w)$

- linear model $\hat{V}(s,w) = W^T X(s)$ (linear)

S: wealthiness
X(s): saving > $$
有 job / not etc.

$X(s)$ [feature construction] < polynomial → $X_i(s) \in [S_1, S_2, S_1^2, S_2^2, S_1 S_2 \dots]$
fourier → $X_i(s) = \cos i\pi S$
radial → $X_i(s) = \exp\left(\frac{\|S - C_i\|}{-2\sigma^2}\right)$

- nonlinear model $\hat{V}(s,w)$ → ANN

- average reward (for continuous task)

$G_t = r_{t+1} + \gamma r_{t+2} + \dots \gamma^{T-1} r_T$, → $G_t = r_{t+1} - \bar{r}_{t+1} + r_{t+2} - \bar{r}_{t+2} + \dots r_T - \bar{r}_T$ < ① × γ, ② differential term

$\bar{r}_{t+1} = \sum_s \mu(s) \sum_a \pi(a|s) \sum_{s', r} p(s', r|s,a)\, r$

$G_t = r + \gamma r + \gamma^2 r + \gamma^3 r \dots = \frac{(1-\gamma^\infty)}{1-\gamma} r \to \frac{1}{1-\gamma}$

- differential term version
  - Bellman equation $V_\pi(s) = \sum_a \pi(a|s) \sum_{s'} p(s', r|s,a)[r + \gamma V_\pi(s')] = \sum\pi(a|s)\sum p(s', r|s,a)[r - \bar{r} + V_\pi(s')]$ ($r \to r - \bar{r}$)
  - TD error $\delta_t = r_{t+1} + \gamma V(s') - V(s) = r_{t+1} - \bar{r}_{t+1} + \hat{V}(S_{t+1}) - \hat{V}(S_t)$ ($r_{t+1} \to r_{t+1} - \bar{r}_{t+1}$; approx)

  for gradient updates $W_t = W_t + \alpha\delta_t\nabla\hat{V}(S_t, W_t)$

- policy gradient

$\pi(a|s) \to \hat{\pi}(a|s, \theta)$

obj func $J(\theta) = \arg\max_\theta V_\pi(S_0)$ → $\nabla J(\theta)$?

- proof of policy gradient theorem:

$\nabla V_\pi(s) = \nabla(\sum_a \pi(a|s)\, q_\pi(s,a,\theta)) = \sum_a[\nabla\pi(a|s)\, q(s,a,\theta) + \pi(a|s)\nabla q_\pi(s,a,\theta)]$

$= \sum_a[\nabla\pi(a|s)\, q(s,a,\theta) + \pi(a|s)\nabla[\sum_{s'} p(s', r|s,a)(r + V_\pi(s'))]]$ ($\nabla\sum(pr + pV) \to \sum\nabla(pr + pV) \to \sum p\nabla V$)

$= \sum_a[\nabla\pi(a|s)\, q(s,a,\theta) + \pi(a|s)\sum_{s'} p(s', r|s,a)\nabla V_\pi(s')$

$= \sum_a[\nabla\pi(a|s)\, q(s,a,\theta) + \pi(a|s)\sum p(s', r|s,a)[\sum_{a'}[\nabla\pi(a'|s')\, q(s', a', \theta) + \pi(a'|s')\sum_{s''} p(s'', r|s', a')\nabla V_\pi(s'')]]$

$= \sum_{x\in S}\sum_k p(s \to x, k, \pi) \times \sum_a \nabla\pi(a|s)\, q(s,a,\theta)$

$\nabla J(\theta) = \nabla V_\pi(S_0) = \sum_s\sum_k p(S_0 \to s, k, \pi)\sum_a\nabla\pi(a|s)\, q(s,a,\theta) = \sum\mu(s)\sum\nabla\pi(a|s)\, q(s,a,\theta) = E_\pi[\sum_a q(s,a,\theta)\nabla\pi(a|s,\theta)] = E[\sum_a q(s,a,\theta)\pi(a|s,\theta)\frac{\nabla\pi}{\pi}] = E_\pi[q(s,A,\theta)\frac{\nabla\pi}{\pi}]$ ($\mu(s)$)

∵ $q(s,A) = E(G_t \mid S, A)$

$= E_\pi[G_t\frac{\nabla\pi}{\pi}]$

$= E_\pi[G_t\nabla\ln\pi(a|s,\theta)]$

$E[\frac{\pi(a|s,\theta)}{b(a|s)} G_t\nabla\ln\pi(a|s,\theta)]$

update $\theta = \theta + \alpha\nabla J(\theta) = \theta + \alpha(G_t\nabla\ln\pi)$ ($\frac{\pi}{b}$) GD → SGD

- REINFORCE [MC + policy gradient]

$(G_t - \text{baseline})$ bias =, var↓ → update $\theta_t = \theta_t + \alpha((G_t - b(S_t))\nabla\ln\pi(S_t, a_t, \theta_t))$ (δ_t → w baseline)
(baseline → any func except func of a)

$b(S_t) = \hat{V}(S_t)$

$r + \hat{V}(s', w)$, $\hat{V}(s,w)$

- AC [TD learning + policy gradient w baseline] ($\frac{\pi}{b}$)

{ update $\theta = \theta + \alpha\nabla J(\theta) = \theta + \alpha[r + \hat{V}_\pi(s', w) - \hat{V}_\pi(s,w)]\nabla\ln\pi(s, a, \theta)$ → policy updates (δ_t = adv term)
{ update $W = W + \alpha\nabla J(w) = w + \alpha[r + \hat{V}(s', w) - \hat{V}(s,w)]\nabla\hat{V}(s,w)$ → value updates

for off-policy: $\theta = \theta + \alpha\frac{\pi}{b}\delta_t\nabla\ln\pi(a|s,\theta)$
$W = W + \alpha\frac{\pi}{b}\delta_t\nabla\hat{V}(s,w)$

- PG < TRPO, PPO, GRPO

$\nabla J(\theta) = E[G_t\nabla\ln\pi(a|s,\theta)] = E[\hat{A}_t\nabla\ln\pi(a|s,\theta)]$ (adv term)

$J(\theta) = E[\hat{A}_t \ln\pi(a|s,\theta)]$

- TRPO $J(\theta) = \arg\max_\theta E[\frac{\pi}{\pi_{old}}\hat{A}_t - \beta KL(\pi \| \pi_{old})]$

- PPO $J^{clip}(\theta) = \arg\max_\theta E[\min(\frac{\pi}{\pi_{old}}\hat{A}_t, \text{clip}(1-\varepsilon, \frac{\pi}{\pi_{old}}, 1+\varepsilon)\hat{A}_t)]$ RLHF

- reward model – r
- actor-critic < policy model – π, critic model – V
- reference model – π_{old}

- GRPO ① reward → rule-based reward
② × critic model $A_t = \frac{r_t - \bar{r}}{\sigma}$

prompt < exp1 → r_1, exp2 → r_2, … exp10 → r_{10}

- SARSA (state-action-reward-state-action)
  - initialize π, $\overset{+b}{}$ $q(s,a)$ for $s \in S$, $a \in A$, $q(S_T, A) = 0$
  - for every episode:

```
initialize $S_0$
$\underline{\overset{b}{\pi}(\cdot|S_0)} \underline{\rightarrow a_0} \rightarrow$ ($\epsilon$-greedy considered)
for t in 0, 1, 2 ... T-1:
    take action $a_t \rightarrow \underline{r_{t+1}, S_{t+1}}$        wait unit n-step to update $q(S_t, a_t)$
    $\overset{b}{\pi}(\cdot|S_{t+1}) \rightarrow a_{t+1}$
    $\underline{q(S_t, a_t)} \leftarrow q(S_t, a_t) + \alpha[r_{t+1} + \underline{\gamma q(S_{t+1}, a_{t+1})} - q(S_t, a_t)]$ → ≈ value updates
    $a^* = \arg\max_a q(S_t, a_t)$
    update $\pi_{a=a^*}(a_t|S_t)$ based on $\epsilon$-greedy    ≻→ ≈ policy updates
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- Q-learning
  - initialize π, $q(s,a)$ for $s \in S$, $a \in A$. $q(S_T, A) = 0$
  - for every episode:

```
initialize $S_0$
$\pi(\cdot|S_0)$ w greedy $\rightarrow a_0$
for t in 0, 1, 2 ... T-1:
    take action $a_t \rightarrow r_{t+1}, S_{t+1}$
    $\pi(\cdot|S_{t+1}) \rightarrow a_{t+1}$
    $q(S_t, a_t) \leftarrow q(S_t, a_t) + \alpha[r_{t+1} + \max_a q(S_{t+1}, a_{t+1}) - q(S_t, a_t)]$
    update $\pi_{a=a^*}(a_t|S_t)$ based on $\epsilon$-greedy
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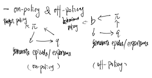
- n-step learning.

$$q(S_t, a_t) = r_{t+1} + \gamma r_{t+2} + \cdots \gamma^{n-1} r_{t+n} + \gamma^n q(S_{t+n}, a_{t+n}) - q(S_t, a_t)$$

- on-policy & off-policy



- importance sampling

$$S_0 \xrightarrow[r_1]{a_0} S_1 \xrightarrow[r_2]{a_1} S_2 \cdots S_T \qquad (S_0, a_0, r_1, S_1, a_1, \cdots)$$

$$\frac{\prod_{t=0}^{T} \pi(a_t|S_t)\, p(S_{t+1}|\cancel{S_t}, a_t)}{\prod_{t=0}^{T} b(a_t|S_t)\, p(S_{t+1}|\cancel{S_t}, a_t)} = \frac{\prod_{0}^{T} \pi(a_t|S_t)}{\prod_{0}^{T} b(a_t|S_t)} = \rho \text{ (importance sampling ratio)}$$